MIL-T-83958A
17 January 1977
SUPERSEDING
MIL-T-83958
5 JANUARY 1971

MILITARY SPECIFICATION

TOWERS, SELF SUPPORTING, TYPE GP-1

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nine sizes of towers, hereinafter referred to as the tower.

1.2 Classification. The towers shall be classified as follows:

a.	GP-1-20	20 feet high (see 3.2.1)
b.	GP-1-30	30 feet high (see 3.2.2)
c.	GP-1-40	40 feet high (see 3.2.3)
d.	GP-1-50	50 feet high (see 3.2.4)
e.	GP-1-60	60 feet high (see 3.2.5)
f.	GP-1-70	70 feet high (see 3.2.6)
g.	GP-1-80	80 feet high (see 3.2.7)
h.	GP-1-90	90 feet high (see 3.2.8)
i.	GP-1-100	100 feet high (see 3.2.9)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

FF-B-575	Bolts, Hexagon and Square
* RR-G-661	Grating, Metal (Floor, Except for Naval Vessels)
* FF-N-836	Nut, Square Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Warner Robins ALC/ MMIRDA, Robins AFB GA 31098, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Military

MIL-P-116	Preservation, Methods of
* MIL-R-11461	Rod, Ground (With Attachments)
MIL-W-27076	Workmanship Standards for Ground Electronic and Associated Equipment

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U S Military Property
MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
* MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment: Requirements for
* MIL-STD-889	Dissimilar Metals

DRAWINGS

United States Air Force

E7032680
LDBWS00534AD000 Sheets 1 and 2
LDBWS00535AD000

Federal Aviation Agency

5439-7	5439-14
5439-8	5439-15
5439-9	5439-16
5439-10	5439-17
5439-11	5439-18

Microflect Drawings

C-16608
D-16607
D-16689

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING MATERIAL

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A-36	Specification for Structural Steel
A-123	Zinc (Hot-Dip Galvanized) Coating on Products Fabricated from Rolled, Pressed, and Forged Steel Shapes, Plates, Bars and Strips
A-153	Zinc Coating (Hot-Dip) on Iron and Steel Hardware

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

* AMERICAN WELDING SOCIETY

Welding Code D1.1

(Applications for copies should be addressed to the American Welding Society, Inc. (AWS), 2501 NW 7th Street, Miami, FL 33125.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 S Riverside Plaza, Chicago, IL 60606.)

AMERICAN TRUCKING ASSOCIATION, INC. (ATA)

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc. (ATA), Traffic Dept, 1616 P Street, NW, Washington, DC 20036.)

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

Occupational Safety and Health Act (OSHA) Standards with Supplemental Revisions

(Applications for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Preproduction testing. This specification makes provision for preproduction testing. (See 4.4)

3.2 Components. The towers covered by this specification shall be composed of various combinations of base section, 20 and/or 10 foot tower sections, a platform and a ladder with safety rail.

Reference drawings:

USAF LDBWS00534AD000, Sheets 1 and 2

The proper combinations of sections and the drawing number for these sections are as follows:

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3.2.1 GP-1-20.

1	Base Section	D-5439-14	Tower Base Section - BAV
2	10 Foot Sections	D-5439-7	Tower Section - E
1	Steel Platform	E7032680	Platform Assembly, Antenna
	Grating	C-16608	
		D-16607	
		D-16689	

3.2.2 GP-1-30.

1	Base Section	D-5439-15	Tower Base Section - SBA - 1
1	20 Foot Section	D-5439-8	Tower Section - S1
1	10 Foot Section	D-5439-7	Tower Section - E
1	Steel Platform	E7032680	Platform Assembly, Antenna
	Grating	C-16608	
		D-16607	
		D-16689	

3.2.3 GP-1-40.

1	Base Section	D-5439-15	Tower Base Section - SBA - 1
1	20 Foot Section	D-5439-8	Tower Section - S1
2	10 Foot Sections	D-5439-7	Tower Section - E
1	Steel Platform	E7032680	Platform Assembly, Antenna
	Grating	C-16608	
		D-16607	
		D-16689	

3.2.4 GP-1-50.

1	Base Section	D-5439-16	Tower Base Section - SBA - 2
1	20 Foot Section	D-5439-9	Tower Section - S2
1	20 Foot Section	D-5439-8	Tower Section - S1
1	10 Foot Section	D-5439-7	Tower Section - E
1	Steel Platform	E7032680	Platform Assembly, Antenna
	Grating	C-16608	
		D-16607	
		D-16689	

3.2.5 GP-1-60.

1	Base Section	D-5439-16	Tower Base Section - SBA - 2
1	20 Foot Section	D-5439-9	Tower Section - S2
1	20 Foot Section	D-5439-8	Tower Section - S1
2	10 Foot Section	D-5439-7	Tower Section - E
1	Steel Platform	E7032680	Platform Assembly, Antenna
	Grating	C-16608	
		D-16607	
		D-16689	

3.2.6 GP-1-70.

1	Base Section	D-5439-17	Tower Base Section - SBA - 3

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1	20 Foot Section	D-5439-10	Tower Section - S3
1	20 Foot Section	D-5439-9	Tower Section - S2
1	20 Foot Section	D-5439-8	Tower Section - S1
1	10 Foot Section	D-5439-7	Tower Section - E
1	Steel Platform Grating	E7032680 C-16608 D-16607 D-16689	Platform Assembly, Antenna

3.2.7 GP-1-80.

1	Base Section	D-5439-17	Tower Base Section - SBA - 3
1	20 Foot Section	D-5439-10	Tower Section - S3
1	20 Foot Section	D-5439-9	Tower Section - S2
1	20 Foot Section	D-5439-8	Tower Section - S1
2	10 Foot Sections	D-5439-7	Tower Section - E
1	Steel Platform Grating	E7032680 C-16608 D-16607 D-16689	Platform Assembly, Antenna

3.2.8 GP-1-90.

1	Base Section	D-5439-18	Tower Base Section - SBA - 4
1	20 Foot Section	D-5439-11	Tower Section - S4
1	20 Foot Section	D-5439-10	Tower Section - S3
1	20 Foot Section	D-5439-9	Tower Section - S2
1	20 Foot Section	D-5439-8	Tower Section - S1
1	10 Foot Section	D-5439-7	Tower Section - E
1	Steel Platform Grating	E7032680 C-16608 D-16607 D-16689	Platform Assembly, Antenna

3.2.9 GP-1-100.

1	Base Section	D-5439-18	Tower Base Section - SBA - 4
1	20 Foot Section	D-5439-11	Tower Section - S4
1	20 Foot Section	D-5439-10	Tower Section - S3
1	20 Foot Section	D-5439-9	Tower Section - S2
1	20 Foot Section	D-5439-8	Tower Section - S1
2	10 Foot Sections	D-5439-7	Tower Section - E
1	Steel Platform Grating	E7032680 C-16608 D-16689 D-16607	Platform Assembly, Antenna

3.3 Materials, parts, and processes. Materials, parts, and processes shall be as specified herein. Materials and parts not covered by applicable specifications shall be of commercial quality, the lightest practical weight, entirely suited for its purpose, and readily available.

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3.3.1 Structural steel. All structural steel shapes such as angles, channels, bars, grating, gussets, flanges, and plates shall be structural steel in accordance with ASTM Standard A-36.

* 3.3.2 Fasteners. Bolts shall be in accordance with FF-B-575, Type 2, Grade 5. Nuts shall be in accordance with FF-N-836, Type 2, Style 4, Grade 1.

* 3.3.3 Grating. The grating to be used for the platform floor shall be steel bar type in accordance with RR-G-661, Type I, and shall be fabricated in accordance with Microflect drawings: C-16608, D-16607 and D-16689.

* 3.3.4 Ground rod. Two 5/8" X 8'-0" ground rods with connecting conductors in accordance with MIL-R-11461 shall be furnished with each tower.

* 3.3.5 Ladder and safety rails.

* 3.3.5.1 Ladder. A steel ladder shall be furnished with each tower by the supplier. The ladder shall meet the requirements of OSHA Standard, paragraph 1910.27 Fixed Ladders.

* 3.3.5.2 Safety rails. Safety rails to be mounted on the ladder shall be supplied with the tower. Safety rails shall have provisions to receive a safety belt clevis which immediately locks upon taking the suspended weight of a man. The safety rail shall meet the requirements of OSHA Standard, paragraph 1910.27(d)(5) Ladder Safety Devices. All necessary hardware shall be provided with the safety rail so that it may be mounted to the rungs of the ladder in accordance with the manufacturer's instructions. At least two leather belts with the required sleeves and clevises for two complete sets of safety belts shall be furnished with each tower. The safety system, including safety rail and safety belts, shall be designed to support a 200 pound man with a factor of safety of 3.

3.4 Zinc coating. All ferrous metal parts shall be zinc-coated (hot dip, galvanized) in accordance with ASTM A-123 for structural steel and ASTM A-153 for hardware.

3.5 Description, general. The towers covered by this specification are four-sided, self-supporting steel structures of various sizes. The towers shall be constructed of galvanized steel, primarily of steel angle, in accordance with referenced drawings. (See 6.3).

* 3.6 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals, as defined by MIL-STD-889, shall not be used in intimate contact.

* 3.7 Workmanship. All members and parts of the tower shall be in accordance with this specification. All shearing, punching, back and end cuts shall be accurately and carefully performed. Holes shall be true and square with the plane of the material and all exposed raw edges, including edges exposed in the dismantled stage shall be ground, machined or otherwise smoothed to prevent accidental cutting or scratching of handlers and users of the tower. The requirements of MIL-W-27076 shall be the primary criteria for establishing minimum acceptable standards of workmanship.

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* 3.8 Welding. All welding shall be done in accordance with the American Welding Society Structural Welding Code D1.1 and all welders shall be certified in accordance with this code. There shall be no welding required for field erection.

3.9 Marking. Identification marking shall be in accordance with MIL-STD-130.

3.9.1 Identification of members. All structural members shall be marked so as to permit ready identification by reference to erection and assembly drawings. Identification marking shall be accomplished by depth stamping. The depth stamping shall be completed prior to finishing, and shall be clearly legible after finishing. The depth stamping shall in no way flatten or deform the round cross-section of tubular members.

3.10 Foundation. The foundation(s) for the tower(s) is (are) not included in this specification. Drawing LDBWS00535AD000 is included for reference only.

3.11 Navigational hazard lights. Navigational hazard lights, photo-electric control, conduit, cabling, etc., required for the installation and operation of navigational hazard lights are not covered by the specification.

3.12 Extra material. An additional 5% of all bolts, nuts, washers and other fasteners shall be furnished. This additional material shall be of the same type and character supplied for the assembly of the tower in the field.

* 3.13 Category III and IV hazards. All equipment furnished under this specification shall be analyzed for Category III and IV hazards as defined by MIL-STD-882. Any hazard so identified shall be brought to the attention of the procuring activity with recommended action for satisfying safety requirements. Action order of precedence shall be as specified in MIL-STD-882, paragraph 5.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. Tests shall be classified as preproduction (see 4.4) and acceptance (see 4.5).

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4.3 Test conditions. Unless otherwise specified, all tests required by this specification shall be made at prevailing ambient temperature and humidity conditions.

4.4 Preproduction test. One tower, or towers, representative of production equipment, and various components as required, shall be subjected to the tests specified herein. The samples shall be identified with the manufacturer's part number and such other information as is required by this specification.

4.4.1 Pre-erection inspection tests. The tower shall be given a thorough mechanical and visual inspection and test to determine that the quality of all materials and workmanship are in compliance with the requirements of this specification. Particular attention shall be given to the following:

a. Choice and quality of materials.
b. Nameplates, identification markings and labels.
c. Finishes.
d. Welded joints.
e. Check of mounting means.
f. Check of corrosion prevention.
g. Check of safety features.
h. Loose fastening and securing devices or parts.
i. Grounding connections.
j. Over-all dimensions check.
k. Other visual defects.
* l. Workmanship (See 3.7).

4.4.2 Assembly. The contractor shall assemble one complete tower of each class specified to insure a proper fit of all members.

4.4.2.1 Post-erection inspection tests. An erected tower shall be subjected to a careful mechanical inspection to insure that the requirements of this specification have been met. Such inspection shall demonstrate the accuracy of fit and the correctness of dimensional data. All sections shall fit easily without having to be forced and without bending or straining of any part. Particular attention shall be given to the following points:

a. The fit of members and parts in their respective positions.
b. Accessibility of parts for servicing and inspection.
c. Check of mounting means.
d. Check of safety features.
e. Nameplate.
f. Correctness of dimensional data.
g. Accuracy of shearing, punching, back and end cuts.

4.5 Acceptance tests. The acceptance tests shall consist of those tests listed in 4.4.1.

4.5.1 Preparation for delivery inspection. The preservation, packaging, packing and marking of the tower shall be inspected to verify conformance to the requirements of Section 5 of this specification.

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5. PREPARATION FOR DELIVERY

5.1 Level of preservation and packaging. The preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Methods of preservation and packaging for tower components and accessories shall be determined and accomplished in accordance with MIL-P-116 and MIL-STD-794. Interior containers shall be determined and utilized in accordance with MIL-STD-794. The contents of each package shall contain items and quantities, which will facilitate efficient and expeditious placement at the erection site for timely unpacking and erection in established steps of succession called for in the erection manual.

5.1.1.2 Fasteners. Bolts, nuts, washers, clips, screws, and similar items of like description shall be preserved in accordance with MIL-P-116, Method IC-2, except where small quantities (2 pounds or less) are involved, then Method IC-3 may be used.

5.1.1.3 Drawings and technical publications. The drawings and technical publications required for each tower shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

5.1.2 Level C. The complete tower shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. The packing shall be A or C, as specified (see 6.2).

5.2.1 Level A. Packing and exterior shipping containers shall conform to the requirements of MIL-STD-794, except that large bar (framing) sheet, plate and structural shape components shall be packed in accordance with MIL-STD-163. The final arrangement of shipping containers shall be in accordance with MIL-STD-794 and lifts conforming to MIL-STD-163 shall be entirely suitable for identification, documentation and accountability of the packages belonging to one(1) tower, when multiple quantities of like towers are being processed or handled simultaneously through shipping and receiving ports.

5.2.2 Level C. The complete tower shall be packed in a manner to insure arrival at destination in satisfactory condition and which will be acceptable to carriers at lowest rates. Packing shall comply with UCC Uniform Freight Classification Rules or National Motor Freight Classification Rules, according to the mode of transportation.

5.3 Marking. Interior packages, bundles, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The GP-1 tower is intended to support air to ground UHF/VHF antennas, AS-1181 and AS-1097.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Class and size of tower required (see 1.2).
c. Level of preservation and packaging (see 5.2).
d. Level of packing (see 5.3).

* 6.3 Drawing changes. In the event it becomes necessary to change the drawings (see 3.5) in order to achieve the performance requirements at this specification, the contractor will immediately notify the contracting officer with recommendations for drawing changes to obtain the required performance. To the extent approved, such changes will be processed in accordance with the provisions of the changes clause of the contract.

6.4 Asterisks. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions, etc.) from the previous issues were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodian:
Air Force - 99

Review Activity:
Army - ME

Using Activities:
Navy - MC, OS, AS

Preparing Activity:
Air Force - 99

Project Number:
5445-0015

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-T-83958A - Towers, Self Supporting, Type GP-1

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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